

3/15

Final Exam

Sciences Lecture Hall 123

Monday March 19th 1:00pm

Topics 11.1 Linear Systems

11.2.1 Compartment Models

11.3 Nonlinear Systems

11.4.4 A Mathematical Model for Neuron Activity

12.1 Counting

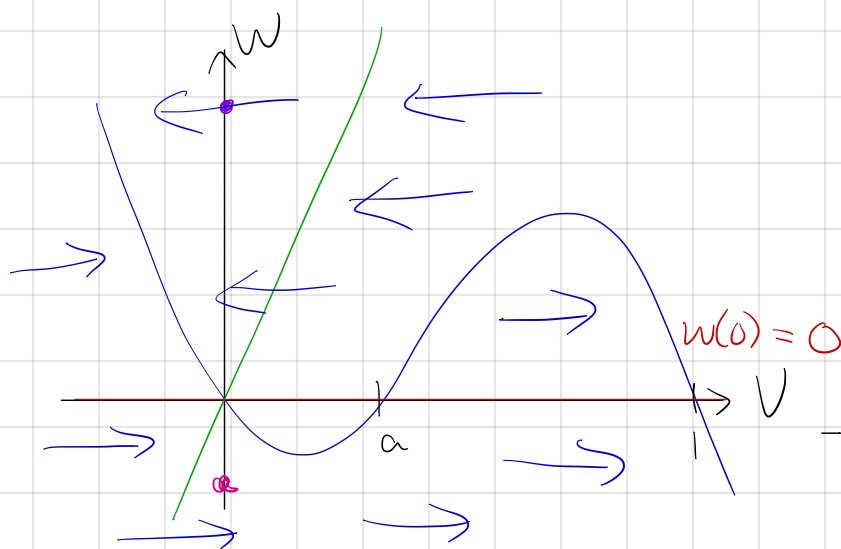
12.2 Probability

12.3.1 Conditional Probability

11.4.4

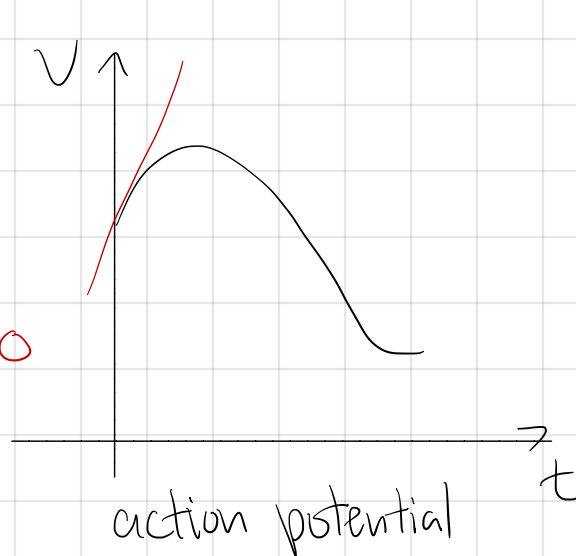
$$\frac{dV}{dt} = -V(V-a)(V-1) - w = 0 \Rightarrow \underline{w = -V(V-a)(V-1)}$$

$$\frac{dw}{dt} = b(V - cw) = 0 \Rightarrow \underline{w = \frac{1}{c}V}$$



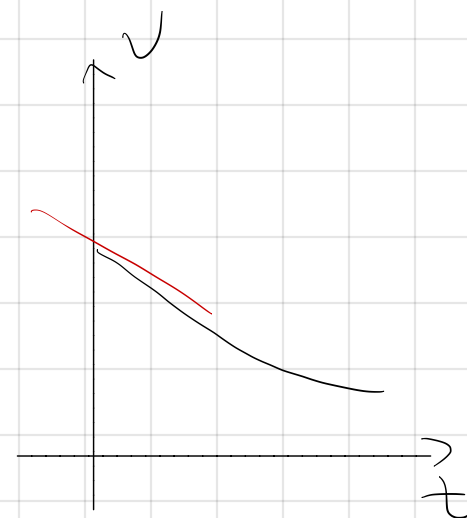
$$(w(0) > 0, V=0) \quad \frac{dV}{dt} = -w < 0$$

$$(w(0) < 0, V=0) \quad \frac{dV}{dt} = -w > 0$$



$$\frac{dV}{dt}(0) > 0$$

$$a < V(0) < 1$$



$$\frac{dV}{dt} < 0$$

$$0 < V(0) < a$$

1. What is the probability of drawing exactly two pairs after drawing 5 cards without replacement?

1 2 1 2

$$\binom{13}{2} \cdot \binom{5}{4} \cdot \binom{44}{1}$$

↑
ways to
choose 2
pairs

↑
ways to put
pairs in
hand

←
choice of
last card

$$\frac{\binom{13}{2} \cdot \binom{5}{4} \cdot \binom{44}{1}}{\binom{52}{5}}$$

↑
all ways to choose
5 cards

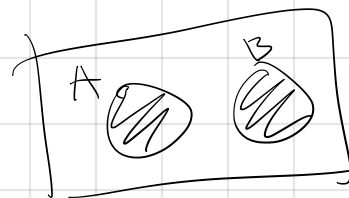
$${}_{52}P_5 = P(52, 5)$$

$${}_{52}C_5 = C(52, 5) = \binom{52}{5}$$

$$2.a) 4 \cdot \binom{10}{5} = |A|$$



$$b) \binom{40}{5} - 4 \cdot \binom{10}{5}$$



$$P(A \cup B) = P(A) + P(B)$$

p.670

Ω

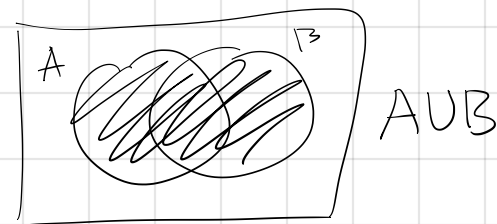
" $|A^c|$ "

$$A^c = \Omega \setminus A$$

$$1) A \subseteq \Omega \quad 0 \leq P(A) \leq 1$$

$$2) P(\emptyset) = 0, P(\Omega) = 1$$

3) disjoint events A, B



$$P(A \cup B) = P(A) + P(B)$$

$$|\Omega| = \binom{40}{5} \quad P(\Omega) = 1 \quad P(\Omega) = P(A^c) + P(A)$$

" $P(A \cup A^c)$ "

1. Toss a coin 3 times

HHH

HHT

HTH

HTT

THH

THT

TTH

TTT

2 · 2 · 2

$$\Omega = \{HHH, HHT, \dots, TTT\}$$

$$|\Omega| = 2 \cdot 2 \cdot 2 = 8$$

12.3 Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

A = 3rd coin is heads

B = 1st coin is heads

$$P(B) = \frac{4}{8} = \frac{1}{2}$$

$$P(A \cap B) = \frac{2}{8} = \frac{1}{4} \quad \frac{\frac{1}{4}}{\frac{1}{2}} = \frac{1}{2}$$